

Relaxation gap of a trajectory-averaged fermionic reset circuit

Class-A fermionic adaptive-circuit project
(Dated: October 1, 2026)

We relate the relaxation of the outcome-averaged density matrix to the single-particle projector product for one measurement cycle. Two-point closure gives an upper bound on the many-body gap. For perfect occupation resets, higher-order neutral correlations cannot relax more slowly, so this bound is saturated. The result applies to a fixed repeated schedule, including noncommuting, support-truncated Wannier measurements.

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I. MEASUREMENT CYCLE AND TWO-POINT RELAXATION

We consider the physical layer after discarding the measurement outcomes and fresh ancillary modes. As in the channel construction of Bhuiyan, Pan, and Jian (BPJ), Appendix D [1], lower-band overcomplete Wannier (OW) modes are filled and upper-band modes emptied. Here we retain the exact discrete resets in their prescribed order, without a continuous-time approximation.

Let $\{\psi_i, \psi_j^\dagger\} = \delta_{ij}$ for N fermion modes. At step j , measure $\hat{\chi}_j = \sum_i v_{ji} \psi_i$, with $v_j^\dagger v_j = 1$ and $\hat{N}_j = \hat{\chi}_j^\dagger \hat{\chi}_j$. Averaging the measurement and perfect feedback gives the physical-layer channels

$$\begin{aligned}\tilde{\mathcal{E}}_{j,0}(\hat{\rho}) &= \hat{\chi}_j \hat{\rho} \hat{\chi}_j^\dagger + (\hat{\mathbf{1}} - \hat{N}_j) \hat{\rho} (\hat{\mathbf{1}} - \hat{N}_j), \\ \tilde{\mathcal{E}}_{j,1}(\hat{\rho}) &= \hat{\chi}_j^\dagger \hat{\rho} \hat{\chi}_j + \hat{N}_j \hat{\rho} \hat{N}_j.\end{aligned}\quad (1)$$

The target occupation is $s_j = 0$ or 1 . The projector terms include measurement dephasing; no separate dephasing step is added. One cycle is the composition $\tilde{\mathcal{E}} = \tilde{\mathcal{E}}_{M,s_M} \circ \dots \circ \tilde{\mathcal{E}}_{1,s_1}$, with step 1 first. Distinct OW modes need not be orthogonal, so this order matters.

Write $\bar{G}_{ab} = \text{Tr}(\bar{\rho} \psi_a^\dagger \psi_b)$ and set $P_j = v_j v_j^\dagger$, $Q_j = \mathbf{1} - P_j$. A reset removes correlations with the measured mode and sets its occupation:

$$\bar{G}' = Q_j \bar{G} Q_j + s_j P_j. \quad (2)$$

The part of the initial two-point function surviving one cycle is therefore propagated by

$$A = Q_M \dots Q_1, \quad \bar{G}_{n+1} = A \bar{G}_n A^\dagger + B. \quad (3)$$

The source B depends on the reset targets, not on the input state. For deviations from a stationary correlation

matrix, $\delta G_{n+1} = A \delta G_n A^\dagger$. Column vectorization gives $A^* \otimes A$, whose eigenvalues are $a_i a_j^*$ if a_i are those of A . Thus, for $0 < r = \rho(A) < 1$,

$$\Delta_{\text{sp}} = -\log r, \quad \Delta_C = -2 \log r = 2 \Delta_{\text{sp}}. \quad (4)$$

All rates are per complete cycle. Here A is the homogeneous correlation propagator, not a single-record transfer matrix. Its spectral radius, rather than its largest singular value, sets the asymptotic two-point decay rate.

II. FROM TWO-POINT DECAY TO THE MANY-BODY GAP

First consider Eq. (3) alone. The Heisenberg channel preserves the span of $\hat{\mathbf{1}}$ and all $\psi_a^\dagger \psi_b$, so its restriction has eigenvalues 1 and $a_i a_j^*$. These also belong to the full channel spectrum. For a mixing channel in any sector containing these operators, the largest nonstationary eigenvalue modulus therefore satisfies

$$r_{\text{MB}} \geq r^2, \quad \boxed{\Delta_{\text{MB}} \leq \Delta_C = 2 \Delta_{\text{sp}}}. \quad (5)$$

This is the spectral-inclusion bound: a positive two-point rate alone cannot exclude a slower higher-order correlation.

To see the slow density mode directly, choose $A^\dagger u = a^* u$. The centered occupation

$$\hat{O}_u = \sum_{ab} u_a^* u_b (\psi_a^\dagger \psi_b - (G_{ss})_{ab} \hat{\mathbf{1}}) \quad (6)$$

obeys $\tilde{\mathcal{E}}^\dagger(\hat{O}_u) = |a|^2 \hat{O}_u$ by Eq. (3). Taking $|a| = r$ constructs a neutral many-body eigenoperator with the two-point decay rate.

For perfect resets we can examine those correlations explicitly. Choose a mode basis starting with $\hat{\chi}_j$, and let $\hat{B}$ contain only orthogonal modes. Equation (1) gives, for an even total operator,

$$\begin{aligned}\tilde{\mathcal{E}}_{j,s}^\dagger(\hat{B}) &= \hat{B} & (\hat{B} \text{ even}), \\ \tilde{\mathcal{E}}_{j,s}^\dagger(\hat{\chi}_j^{(\dagger)} \hat{B}) &= 0 & (\hat{B} \text{ odd}), \\ \tilde{\mathcal{E}}_{j,s}^\dagger(\hat{N}_j \hat{B}) &= s \hat{B} & (\hat{B} \text{ even}).\end{aligned} \quad (7)$$

Physically, a reset erases a measured leg of a correlation. A measured occupation pair contributes its fixed value

and leaves a correlation of two fewer fermion operators. A $2p$ -point function therefore couples only to itself and lower-order functions. This operator identity does not assume Wick factorization of the input state.

Charge-neutral correlations take the form

$$F_{I;J}^{(p)} = \text{Tr} \left(\hat{\rho} \hat{\psi}_{i_1}^\dagger \cdots \hat{\psi}_{i_p}^\dagger \hat{\psi}_{j_p} \cdots \hat{\psi}_{j_1} \right), \quad (8)$$

where I, J are increasing p -element index sets. The part remaining of order $2p$ propagates each creation index with A and each annihilation index with A^* . Fermion antisymmetry gives its matrix entries over one cycle:

$$H_p[I, J; K, L] = \det A[I, K] \det A[J, L]^*. \quad (9)$$

The same expression follows step by step from Q_j , composing its minors in the actual order by the Cauchy–Binet identity. Ordering moments by p makes their evolution block triangular; terms coupling different orders do not change its eigenvalues. The multipliers of Eq. (9) are

$$\Lambda_{I,J} = \prod_{i \in I} a_i \prod_{j \in J} a_j^*, \quad |I| = |J| = p. \quad (10)$$

The sets are independent and may overlap: a density fluctuation can use the same single-particle label on both legs. These moments span all neutral operators, with $\sum_p \binom{N}{p}^2 = \binom{2N}{N}$ basis elements. Schur triangularization gives the same products even for a nondiagonalizable A .

The constant moment ($p = 0$) supplies eigenvalue 1. Every $p \geq 1$ product obeys $|\Lambda_{I,J}| \leq r^{2p} \leq r^2$. A density fluctuation attains r^2 at $p = 1$ by choosing $I = J = \{i_*\}$ with $|a_{i_*}| = r$. Thus no higher neutral correlation is slower, the stationary eigenvalue in this sector is simple, and

$$\Delta_{q=0} = \Delta_C = -2 \log \rho(A). \quad (11)$$

III. PHYSICAL SECTOR AND EARLIER RESULTS

Charge neutral means $[\hat{N}_F, \hat{\rho}] = 0$, where $\hat{N}_F = \sum_i \hat{\psi}_i^\dagger \hat{\psi}_i$; it permits mixtures of number sectors. Definite-

charge reset Kraus operators preserve this condition while changing particle number. Maximal mixing and fixed-number Slater preparations both lie in this sector.

With p creations and ℓ annihilations, the even-sector argument gives products of $p + \ell$ factors. Every nonconstant even product has at least two, while the neutral sector attains r^2 . Hence $\Delta_{\text{even}} = \Delta_{q=0}$. To include odd operators, use the similarity $\mathcal{S}(\hat{O}) = (-1)^{\hat{N}_F} \hat{O}$. It changes the sign of the odd-Kraus sandwich in $\mathcal{S} \tilde{\mathcal{E}}_j^\dagger \mathcal{S}$; on odd operators that sign cancels the fermionic reordering sign and restores Eq. (7). Odd products then have maximal modulus r , attained with one factor. Thus $\Delta_{\text{full}} = \Delta_{\text{sp}} = \Delta_C/2$ for the specified Kraus extension; physical density matrices are parity even.

For an explicit continuous-time counterpart, see Barthel–Zhang, Sec. III.6: Proposition 7 and Eq. (63) identify the covariance generator inside the many-body Liouvillian spectrum [2]. Their Sec. III.5, Proposition 6 and the discussion after Eq. (57d), distinguish full and two-point gaps in quasi-free Lindbladians. Equation (5) is the discrete-channel counterpart, derived as Eq. (4) in the September 29 project note [3]. Equality (11) here requires the additional exact-reset argument.

BPJ Appendix D, Eqs. (D34)–(D40), instead estimates relaxation from band dissipation rates in an effective continuous-time, untruncated bulk description [1]; it does not establish the present finite-cycle identity. Our result allows nonnormal transients and Jordan prefactors. It fixes an asymptotic spectral rate, not a size-uniform mixing time or conditional-trajectory purification time. Extra dephasing, imperfect feedback, or averaging different schedules requires a new higher-order analysis.

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- [1] A. Bhuiyan, H. Pan, and C.-M. Jian, *Phys. Rev. Research* **8**, 023147 (2026), Appendix D.
 [2] T. Barthel and Y. Zhang, *J. Stat. Mech.* (2022) 113101, Secs. III.5–III.6; [arXiv:2112.08344v5](#), pp. 14–16.

- [3] A. Bhuiyan, *A spectral bound on relaxation of the trajectory-averaged circuit*, project note (September 29, 2026), Eq. (4); [local PDF](#).